

(TR-104) PYTHON WITH AI/ML

Training Day 75 Report

Date:24 April 2026

Topic Covered: Insurance Data Integration, FastAPI APIs, LangChain Agent Enhancement & Backend Optimization

The seventy-fifth day of training focused on integrating insurance data into MongoDB, developing insurance-related APIs using FastAPI, enhancing LangChain agent capabilities, exploring Excel export functionality, and optimizing backend performance.

- Created and integrated insurance datasets linked with trailers using ObjectId references.
- Developed FastAPI APIs for insurance management, including policy retrieval, status filtering, and expiry tracking.
- Enhanced LangChain agents by integrating insurance-related tools and extending support for policy and renewal queries.
- Expanded the tool-based query system to support trailers, customers, payments, and insurance analytics.
- Implemented advanced MongoDB aggregation pipelines for handling complex business queries.
- Explored Excel export functionality using pandas, openpyxl, BytesIO, and FastAPI StreamingResponse.
- Improved MongoDB document formatting, serialization, caching mechanisms, and query optimization techniques.
- Integrated LangChain agent with FastAPI chat endpoints for dynamic query handling and structured responses.
- Reviewed poster content and verified spelling, grammar, and overall presentation quality.
- Revised concepts related to MongoDB collection integration, LangChain agents, tool-based architectures, aggregation pipelines, FastAPI APIs, and Excel export mechanisms.
- GitHub Link: https://github.com/JaspinderKaurWalia26/practice_project

Overall, the training enhanced understanding of MongoDB data integration, FastAPI API development, LangChain-based AI agents, backend optimization strategies, and scalable data management solutions.